Supplementary Document: Expert Panel & Content Validity Ratio (CVR) Analysis

This document provides the full quantitative breakdown of the Content Validity Ratio (CVR) analysis conducted to establish the content and structural validity of the Presence Factor Scale (PFS), in accordance with Lawshe's methodology (1975) and the Standards for Educational and Psychological Testing.

1. Methodology & Decision Thresholds

An international panel of independent Virtual Reality and presence experts from 7 countries (Australia, Brazil, Canada, France, Germany, the United Kingdom, and the United States) evaluated an initial candidate pool of 38 factors extracted from our Systematic Literature Review (SLR). Each panelist independently rated each factor using a 3-point categorical scale:

- **Essential** (Indispensable for evaluating presence)
- **Useful, but not essential**
- **Not necessary**

The Content Validity Ratio was computed using Lawshe's formula:

$$CVR = \frac{n_e - (N_p/2)}{N_p/2} = \frac{n_e - 4.5}{4.5}$$

where $n_e$represents the number of panelists rating the factor as "Essential" and $N_p = 9$is the total panel size. In accordance with Lawshe (1975), the critical one-tailed significance threshold at $p < 0.05$ for a 9-expert panel is $CVR \geq 0.78$ (requiring at least 8 out of 9 "Essential" ratings).

2. Summary of Findings

- **Retained Objective Factors (23 Factors):** All 23 retained factors achieved statistical significance ($CVR \geq 0.78, p < 0.05$), confirming their role as primary, configurable determinants of spatial presence within the system and environment.
- **Consolidated / Excluded Technical Factors (6 Factors):** Redundant technical parameters failed to achieve individual critical significance and were systematically consolidated into broader primary dimensions (e.g., selection metaphors merged into Interactive Manipulation IF1; thermal/wind cues merged into Haptics SF13).
- **Excluded Subjective User Trait Factors (9 Factors):** All proposed candidate variables related to individual user traits (personality dimensions, immersive tendency, prior gaming proficiency, spatial ability) failed to meet the significance threshold ($CVR \leq 0.33, p > 0.05$). Panelists reached strong consensus that while user traits introduce experiential variance during exposure, they are non-configurable system variables and must be excluded from an objective engineering inspection rubric.

3. Comprehensive Factor Evaluation Matrix ($N_p = 9$ Experts)

ID	Candidate Factor	Initial Category	Essential Votes (n_e)	Computed CVR	Decision ($p < 0.05$)
01	Inclusiveness	Engagement	9 / 9	**+1.00**	**Retained (PFS — E1)**
02	Narrative Structure	Engagement	8 / 9	**+0.78**	**Retained (PFS — E2)**
03	Attention Allocation	Engagement	8 / 9	**+0.78**	**Retained (PFS — E3)**
04	Emotional Elicitation	Engagement	8 / 9	**+0.78**	**Retained (PFS — E4)**
05	Task Complexity	Engagement	8 / 9	**+0.78**	**Retained (PFS — E5)**
06	Copresence	Engagement	8 / 9	**+0.78**	**Retained (PFS — E6)**
07	Social Presence	Engagement	8 / 9	**+0.78**	**Retained (PFS — E7)**
08	Manipulation Intuitiveness	Interactive Fidelity	9 / 9	**+1.00**	**Retained (PFS — IF1)**
09	Travel Intuitiveness	Interactive Fidelity	9 / 9	**+1.00**	**Retained (PFS — IF2)**
10	Selection Metaphor	Interactive Fidelity	5 / 9	+0.11	Excluded (Consolidated into IF1)
11	Invasiveness (Device Burden)	Sensorial Fidelity	8 / 9	**+0.78**	**Retained (PFS — SF1)**
12	Proprioceptive Feedback	Sensorial Fidelity	8 / 9	**+0.78**	**Retained (PFS — SF2)**
13	Body Ownership (Embodiment)	Sensorial Fidelity	9 / 9	**+1.00**	**Retained (PFS — SF3)**
14	Motion-to-Photon Latency	Sensorial Fidelity	9 / 9	**+1.00**	**Retained (PFS — SF4)**
15	Transition Smoothness	Sensorial Fidelity	8 / 9	**+0.78**	**Retained (PFS — SF5)**
16	Multimodal Synchronization	Sensorial Fidelity	9 / 9	**+1.00**	**Retained (PFS — SF6)**
17	Plausibility Illusion	Sensorial Fidelity	9 / 9	**+1.00**	**Retained (PFS — SF7)**
18	Visual Realism	Sensorial Fidelity	8 / 9	**+0.78**	**Retained (PFS — SF8)**
19	Field of View (FOV)	Sensorial Fidelity	9 / 9	**+1.00**	**Retained (PFS — SF9)**

ID	Candidate Factor	Initial Category	Essential Votes (n_e)	Computed CVR	Decision ($p < 0.05$)
20	Display Resolution	Sensorial Fidelity	8 / 9	**+0.78**	**Retained (PFS — SF10)**
21	Frame Rate (FPS)	Sensorial Fidelity	9 / 9	**+1.00**	**Retained (PFS — SF11)**
22	Acoustic Spatialization	Sensorial Fidelity	9 / 9	**+1.00**	**Retained (PFS — SF12)**
23	Haptic Feedback	Sensorial Fidelity	8 / 9	**+0.78**	**Retained (PFS — SF13)**
24	Olfactory Cues	Sensorial Fidelity	8 / 9	**+0.78**	**Retained (PFS — SF14)**
25	Gustatory Feedback	Sensorial Fidelity	2 / 9	-0.56	Excluded (Non-significant)
26	Wind Simulation	Sensorial Fidelity	4 / 9	-0.11	Excluded (Consolidated into SF13)
27	Thermal Feedback	Sensorial Fidelity	4 / 9	-0.11	Excluded (Consolidated into SF13)
28	Dynamic Lighting Contrast	Sensorial Fidelity	5 / 9	+0.11	Excluded (Consolidated into SF8)
29	Stereoscopic Depth Budget	Sensorial Fidelity	6 / 9	+0.33	Excluded (Consolidated into SF9/SF10)
30	Immersive Tendency	Personal Characteristics	5 / 9	+0.11	Excluded (User-Level Variance)
31	Prior Video Game Experience	Personal Characteristics	4 / 9	-0.11	Excluded (User-Level Variance)
32	Spatial Ability / Mental Rotation	Personal Characteristics	5 / 9	+0.11	Excluded (User-Level Variance)
33	Big Five Personality Traits	Personal Characteristics	2 / 9	-0.56	Excluded (User-Level Variance)
34	Age & Demographic Profile	Personal Characteristics	3 / 9	-0.33	Excluded (User-Level Variance)
35	Prior VR Familiarity	Personal Characteristics	5 / 9	+0.11	Excluded (User-Level Variance)
36	Cognitive Absorption Capacity	Personal Characteristics	4 / 9	-0.11	Excluded (User-Level Variance)
37	Susceptibility to Motion Sickness	Personal Characteristics	6 / 9	+0.33	Excluded (User-Level Variance)
38	Domain-Specific Knowledge	Personal Characteristics	3 / 9	-0.33	Excluded (User-Level Variance)